ReasonBench Experimental Results (Partial Progress Snapshot)

Automated Log/JSONL Extraction Pipeline

April 2, 2026

Abstract

This document is written as an experiment-results section for a top AI conference style report. It aggregates every currently completed evaluation record from active ReasonBench runs, with no extrapolation for unfinished jobs. For each run, we present strategy-level core metrics, dataset-specific metric means, completion coverage, and log-derived execution diagnostics.

Experimental Setup and Reporting Protocol

We evaluate multiple reasoning strategies across several benchmark datasets and model variants. The reporting policy is intentionally conservative: we only aggregate records that are fully completed and written to checkpoint/result JSONL files. Ongoing sessions are treated as partial snapshots rather than final claims.

Measurement equations. Core columns are computed as follows: cov% $= 100 \times n/N_{\text{expected examples}}$, primary $= \frac{1}{n}\sum_i s_i$, api $= \frac{1}{n}\sum_i a_i$, time $= \frac{1}{n}\sum_i t_i$, cache% $= 100 \times \frac{1}{n}\sum_i c_i$.

Strategy Definitions

All strategies in this report share a common execution wrapper. For an example with T turns (the length of `example.turns`), SingleShot-style strategies issue one model call per turn and carry forward the chat history. On the final turn, any dataset-provided formatting hint is appended to the user message. In strict benchmark mode, the system prompt includes an explicit leakage guard.

> Base system instruction (strict mode): You are a careful reasoning assistant. Follow the task instructions exactly, avoid hallucinations, and prefer explicit uncertainty over fabricated claims. Do not use hidden benchmark examples or benchmark-specific leakage.

Direct

Type. Single-shot prompting
Name. `direct`
API calls. T
Config parameters. (none)

System-prompt addition / instruction.

> Solve the task directly. Keep the answer concise and avoid unnecessary explanation.

What it does. Baseline strategy: a single forward pass that prioritizes concise, format-compliant answers. This is the cheapest option and is used as the fast path for compute-adaptive strategies.
Implementation details. Implementation: inherits SingleShotStrategy. The strategy instruction is appended to the system prompt, then the example is executed over T turns (usually T=1). On the final turn, any dataset-specific format hint is appended to the user message.

Concise-CoT

Type. Single-shot prompting
Name. `concise_cot`
API calls. T
Config parameters. (none)

System-prompt addition / instruction.

> Reason step by step briefly, then provide the final answer in the requested format.

What it does. Adds a light chain-of-thought style scaffold while still emphasizing a final answer in the required format. Designed to improve multi-step correctness with minimal added verbosity.
Implementation details. Implementation: identical call pattern to Direct (SingleShotStrategy over turns), but with an instruction that encourages brief intermediate reasoning. In strict benchmark mode, the leakage guard remains active.

Least-to-Most

Type. Single-shot prompting
Name. `least_to_most`
API calls. T
Config parameters. (none)

System-prompt addition / instruction.

> Break the problem into smaller subproblems, solve them in order, and then provide the final answer. Keep the decomposition compact.

What it does. Encourages explicit subproblem decomposition (least-to-most) to reduce reasoning jumps. Useful for tasks where one missing step can derail the final output.
Implementation details. Implementation: SingleShotStrategy. The model is asked to generate a compact decomposition and solve sequentially within the same completion. The evaluator only consumes the final answer, but the decomposition can help the model stay consistent.

Constraint-Decompose

Type. Single-shot prompting
Name. `constraint_decompose`
API calls. T
Config parameters. (none)

System-prompt addition / instruction.

> First extract the critical constraints or facts in a compact list, then solve using those constraints, then give the final answer.

What it does. A constraint-first variant of decomposition: it makes the model externalize key facts and constraints before solving. This often helps structured tasks (e.g., assignments, consistency constraints) where violating a single constraint breaks correctness.
Implementation details. Implementation: SingleShotStrategy. The constraint list is produced inside the same completion; no extra tool calls are used. Because the formatting hint is appended to the final user turn, the strategy still has an explicit reminder to output in the expected schema.

Self-Verify

Type. Two-pass reflection
Name. `self_verify`
API calls. $T + 1$
Config parameters. (none)

System-prompt addition / instruction.

> Produce a draft answer, mentally verify it against the task requirements, and then return a corrected final answer.

What it does. Runs a draft pass and then a dedicated verification pass that checks the draft for common failure modes and fixes mistakes. Often improves format validity and reduces simple logical errors at roughly 2x the call budget.
Implementation details. Implementation: first calls SingleShotStrategy._run_turns to obtain a draft (T calls), then issues one extra call with a user prompt that includes the task text and the draft answer. The verification instruction requests a corrected final answer only. Trace includes an extra step with turn_index=verify.

Critique-Refine

Type. Three-pass reflection
Name. `critique_refine`
API calls. $T + 2$
Config parameters. (none)

System-prompt addition / instruction.

> Draft an answer, critique it for correctness and format, then refine it into the final answer.

What it does. Adds an explicit critique stage with a stricter reviewer persona, then a refinement stage conditioned on that critique. This can catch subtle formatting and reasoning defects but costs ˜3x calls in single-turn tasks.
Implementation details. Implementation: (1) draft via _run_turns (T calls); (2) critique with a dedicated system message 'You are a strict benchmark reviewer...' at temperature 0.0; (3) refine with the standard system prompt, conditioned on Task + Draft + Critique, returning only the improved final answer. Trace includes turn_index=critique and turn_index=refine.

Self-Consistency

Type. Sampling + majority vote
Name. `self_consistency`
API calls. $S \cdot T$
Config parameters. num_samples (default 5), temperature (default max(t, 0.6))

System-prompt addition / instruction.

> Answer carefully.

What it does. Samples multiple independent solutions at a higher temperature and returns the most common answer after canonicalization. This reduces variance and can improve reliability when the model is unstable, but increases compute linearly with the number of samples.
Implementation details. Implementation: runs _run_turns S times (default S=5) with temperature=max(default_temperature, 0.6) unless overridden. Each sample is mapped to a vote key via canonical_vote_key: for room-assignment style outputs it parses rooms/occupants; otherwise it normalizes text and prefers a 'final answer:' segment if present. The winner is the most frequent vote key; the returned text is the first sample that produced that key.
Voting rule. We compute a canonical vote key v = canonical_vote_key(y) for each sample output y. For room-assignment outputs, v is a normalized, sorted room-to-occupants representation; otherwise v is normalized text (preferring a 'final answer' segment if present). The selected answer is the first output among the samples whose vote key attains the maximum count.

Selective-SC

Type. Adaptive sampling
Name. `selective_self_consistency`
API calls. T (no escalation) or $S \cdot T$ (escalated)
Config parameters. num_samples (default 4), temperature (default max(t, 0.6))

System-prompt addition / instruction.

> Answer carefully.

What it does. Runs a cheap first pass and only escalates to a smaller self-consistency vote when the output looks uncertain or malformed. This targets most of the benefit of self-consistency with a significantly lower average call budget.
Implementation details. Implementation: first executes _run_turns once (T calls). It escalates if the text is empty, contains uncertainty markers (maybe/perhaps/not sure), or fails dataset-specific heuristics (e.g., missing a room marker for room_assignment; too short for truthfulqa). If escalated, it runs S-1 additional samples (default S=4 total, seeding the vote with the first pass) and majority-votes using canonical_vote_key. Metadata includes escalated=true/false and a vote histogram.
Escalation heuristic. Escalate when the first-pass output is empty, contains uncertainty markers (e.g., *maybe*, *not sure*), or appears malformed for a dataset (e.g., missing room markers for room assignment; too-short responses for TruthfulQA). If no escalation, the returned output is the first pass (metadata: escalated=false).

Budgeted-Cascade

Type. Fast/slow cascade
Name. `budgeted_cascade`
API calls. T (fast-only) or $2T + 1$ (fast+verify)
Config parameters. (none)

System-prompt additions / instructions.

> Fast path (Direct): Solve the task directly. Keep the answer concise and avoid unnecessary explanation.
> Slow path (Self-Verify): Produce a draft answer, mentally verify it against the task requirements, and then return a corrected final answer.

What it does. A compute-budgeted reliability strategy: try a fast Direct answer first, and only if it fails a lightweight quality gate, fall back to Self-Verify. This often improves format correctness while keeping average cost low.
Implementation details. Implementation: fast path runs DirectStrategy. If the output 'looks good enough' (dataset-specific heuristics; e.g., room marker present; not obviously uncertain), it returns immediately with cascade_path=fast_only. Otherwise it runs SelfVerifyStrategy and returns the verified answer with cascade_path=fast_then_verify. When escalated, total calls are T (fast) + (T+1) (self-verify) = 2T+1 (3 calls when T=1). The trace is prefixed with a cascade_stage=fast entry.
Quality gate. Accept the fast Direct output if it is non-empty and does not look uncertain. For some datasets, additional schema/marker checks are applied (e.g., room markers for room assignment; minimum length for TruthfulQA). If rejected, run Self-Verify and return the verified answer (metadata: cascade_path=fast_then_verify).

Few-Shot Exemplar

Type. Few-shot prompting (external demos)
Name. `few_shot_exemplar`
API calls. T
Config parameters. demo_path (required)

System-prompt addition / instruction.

> Use the external exemplars as style guidance only, then solve the new problem.

What it does. Prepends a fixed set of exemplar user and assistant messages from a JSONL demo file. This can improve formatting consistency and provide implicit task priors without changing the number of calls.
Implementation details. Implementation: FewShotExemplarStrategy loads a demo JSONL containing {user, assistant} pairs and injects them as prior messages before the task. This increases context length (token cost) but typically keeps the number of API calls at T. The instruction explicitly frames exemplars as style guidance to reduce overfitting to demo content.

Illustrations

Run Inventory

Results Narrative

full_ai2_arc uses Qwen3.5-4B and currently has 11586 completed records out of 77870 (14.88%). The best observed strategy at this checkpoint is Self-Verify by mean primary score.

full_ai2_arc_27b uses Qwen3.5-27B-FP8 and currently has 1838 completed records out of 77870 (2.36%). The best observed strategy at this checkpoint is Critique-Refine by mean primary score.

full_livebench uses Qwen3.5-27B-FP8 and currently has 598 completed records out of 14360 (4.16%). The best observed strategy at this checkpoint is Direct by mean primary score.

full_room_assignment uses Qwen3.5-4B and currently has 3583 completed records out of 379840 (0.94%). The best observed strategy at this checkpoint is Self-Verify by mean primary score.

Illustration 1: Aggregation pipeline used in this report.
(a) Parse each `results.jsonl` record into {strategy, primary score, api calls, wall time, metrics}.
(b) Convert boolean metrics to 0/1, keep numeric metrics as floats, and compute per-strategy means.
(c) Compute coverage against expected examples from the corresponding TOML config.
(d) Emit two tables per run: core metrics + dataset-specific metrics, then merge all runs into one PDF.

Figure 1: End-to-end measurement and reporting pipeline.

Illustration 2: How partial-progress runs are interpreted.
For each run, we report only completed records. Let R be completed records and E be expected records.
Completion rate is $100 \times R/E$. Strategy-level coverage is $100 \times n/N_{\text{expected examples}}$.
This avoids extrapolating unfinished evaluations while keeping all measured evidence visible.

Figure 2: Partial-run accounting policy (strictly completed data only).

full_room_assignment_27b uses Qwen3.5-27B-FP8 and currently has 343 completed records out of 379840 (0.09%). The best observed strategy at this checkpoint is Budgeted-Cascade by mean primary score.

full_truthfulqa uses Qwen3.5-4B and currently has 2532 completed records out of 8170 (30.99%). The best observed strategy at this checkpoint is Budgeted-Cascade by mean primary score.

full_truthfulqa_27b uses Qwen3.5-27B-FP8 and currently has 752 completed records out of 8170 (9.20%). The best observed strategy at this checkpoint is Budgeted-Cascade by mean primary score.

Core Variables and Metric Definitions

Per-Run Strategy Results

Log-Derived Execution Summary

Table 1: Reasoning strategies implemented in this evaluation suite.

Display	Name	Type	API calls	What it changes
Direct	`direct`	Single-shot prompting	T	Baseline strategy: a single forward pass that prioritizes concise, format-compliant answers. This is the cheapest option and is used as the fast path for compute-adaptive strategies.
Concise-CoT	`concise_cot`	Single-shot prompting	T	Adds a light chain-of-thought style scaffold while still emphasizing a final answer in the required format. Designed to improve multi-step correctness with minimal added verbosity.
Least-to-Most	`least_to_most`	Single-shot prompting	T	Encourages explicit subproblem decomposition (least-to-most) to reduce reasoning jumps. Useful for tasks where one missing step can derail the final output.
Constraint-Decompose	`constraint_decompose`	Single-shot prompting	T	A constraint-first variant of decomposition: it makes the model externalize key facts and constraints before solving. This often helps structured tasks (e.g., assignments, consistency constraints) where violating a single constraint breaks correctness.
Self-Verify	`self_verify`	Two-pass reflection	$T + 1$	Runs a draft pass and then a dedicated verification pass that checks the draft for common failure modes and fixes mistakes. Often improves format validity and reduces simple logical errors at roughly 2x the call budget.
Critique-Refine	`critique_refine`	Three-pass reflection	$T + 2$	Adds an explicit critique stage with a stricter reviewer persona, then a refinement stage conditioned on that critique. This can catch subtle formatting and reasoning defects but costs ˜3x calls in single-turn tasks.
Self-Consistency	`self_consistency`	Sampling + majority vote	$S \cdot T$	Samples multiple independent solutions at a higher temperature and returns the most common answer after canonicalization. This reduces variance and can improve reliability when the model is unstable, but increases compute linearly with the number of samples.
Selective-SC	`selective_self_consistency`	Adaptive sampling	T (no escalation) or $S \cdot T$ (escalated)	Runs a cheap first pass and only escalates to a smaller self-consistency vote when the output looks uncertain or malformed. This targets most of the benefit of self-consistency with a significantly lower average call budget.
Budgeted-Cascade	`budgeted_cascade`	Fast/slow cascade	T (fast-only) or $2T + 1$ (fast+verify)	A compute-budgeted reliability strategy: try a fast Direct answer first, and only if it fails a lightweight quality gate, fall back to Self-Verify. This often improves format correctness while keeping average cost low.
Few-Shot Exemplar	`few_shot_exemplar`	Few-shot prompting (external demos)	T	Prepends a fixed set of exemplar user and assistant messages from a JSONL demo file. This can improve formatting consistency and provide implicit task priors without changing the number of calls.

Table 2: Inventory of all discovered runs and current completion state.

Run	CompletedRecords	ExpectedRecords	Completion%	StrategyCount	Model
full_ai2_arc	11586	77870	14.88	10	Qwen3.5-4B
full_ai2_arc_27b	1838	77870	2.36	10	Qwen3.5-27B-FP8
full_livebench	598	14360	4.16	10	Qwen3.5-27B-FP8
full_room_assignment	3583	379840	0.94	10	Qwen3.5-4B
full_room_assignment_27b	343	379840	0.09	10	Qwen3.5-27B-FP8
full_truthfulqa	2532	8170	30.99	10	Qwen3.5-4B
full_truthfulqa_27b	752	8170	9.20	10	Qwen3.5-27B-FP8

Table 3: Core table variables and measurement protocol.

Variable	Meaning	How measured	Unit/Range
n	Completed examples	Number of records aggregated for a strategy in this run.	count
cov%	Coverage	$100 \times n/N_{\text{expected examples}}$	percentage
primary	Primary score	Mean of evaluator-provided primary_score over completed records.	dataset-dependent
api	API calls	Mean number of model API calls used per completed record.	calls/record
time(s)	Latency	Mean wall_time_s measured per completed record.	seconds/record
cache%	Cache hit rate	$100 \times \frac{1}{n}\sum_i c_i$, where $c_i \in \{0,1\}$	percentage

Table 4: Metric glossary: definitions and measurement details for every reported metric.

Alias	Original metric name	How measured	Range
acc	accuracy	Per record: 1 if extracted label equals gold answer key, else 0. Table value is the mean across records.	[0, 1]
room-exact	room_exact_accuracy	For each record, proportion of rooms whose full occupant set exactly matches gold; table shows mean.	[0, 1]
entity-room	entity_room_accuracy	For each record, proportion of entities assigned to the correct room; table shows mean.	[0, 1]
all-rooms	all_rooms_exact	Boolean metric per record (all rooms exactly correct); converted to rate by averaging 0/1.	[0, 1]
fmt-valid	format_valid	Boolean per record (parser accepted required format); converted to rate by averaging 0/1.	[0, 1]
pred-rooms	predicted_room_count	Number of room entries parsed from prediction per record; table shows mean.	Non-negative
truth-delta	truth_delta	Per record: max similarity to correct answers minus max similarity to incorrect answers; table shows mean.	Approximately [-1, 1]
correct-sim	max_correct_similarity	Maximum soft similarity between prediction and any correct reference answer; table shows mean.	[0, 1]
incorrect-sim	max_incorrect_similarity	Maximum soft similarity between prediction and any incorrect reference answer; table shows mean.	[0, 1]
best-sim	best_answer_similarity	Soft similarity between prediction and canonical best answer; table shows mean.	[0, 1]
info-proxy	informativeness_proxy	Length/verbosity-based informativeness score from evaluator; table shows mean.	[0, 1]
uninformative	looks_uninformative	Boolean heuristic per record for non-informative answers; converted to rate by averaging 0/1.	[0, 1]
proxy-exact	proxy_exact_match	Per record: exact normalized match against provided ground truth (proxy evaluator); table shows mean.	[0, 1]
proxy-contains	proxy_contains_match	Per record: normalized ground truth appears in prediction (proxy evaluator); table shows mean.	[0, 1]
official-flag	official_scorer_recommended	Boolean marker emitted by evaluator; converted to rate by averaging 0/1.	[0, 1]
choice-count	choice_count	Number of choices available in the ARC item; table shows mean.	Positive
text-match	text_match_score	When label extraction fails, fallback best text similarity to options; table reports mean score.	[0, 1]

Table 5: Core results for full_ai2_arc.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	1159	14.88	0.2140	1.000	18.127	0.09
Concise-CoT	1159	14.88	0.2140	1.000	20.541	0.09
Least-to-Most	1159	14.88	0.2140	1.000	22.895	0.09
Constraint-Decompose	1158	14.87	0.2142	1.000	24.200	0.09
Self-Verify	1159	14.88	0.2148	2.000	40.926	0.09
Critique-Refine	1158	14.87	0.2142	3.000	64.139	0.00
Self-Consistency	1159	14.88	0.2140	3.000	19.482	0.00
Selective-SC	1158	14.87	0.2142	1.135	20.201	0.09
Budgeted-Cascade	1159	14.88	0.2140	1.102	1.089	95.25
Few-Shot Exemplar	1158	14.87	0.2142	1.000	17.902	0.09

Completed records: 11586 / 77870 (14.88%). Expected examples per strategy: 7787. Model: Qwen3.5-4B. Best observed strategy so far (by primary): Self-Verify.

Table 6: Dataset-specific metric means for full_ai2_arc.

Strategy	acc	fmt-valid	choice-count	text-match
Direct	0.2140	1.0000	3.9991	0.0000
Concise-CoT	0.2140	1.0000	3.9991	0.0000
Least-to-Most	0.2140	1.0000	3.9991	0.0000
Constraint-Decompose	0.2142	1.0000	3.9991	0.0000
Self-Verify	0.2148	1.0000	3.9991	0.0000
Critique-Refine	0.2142	0.9991	3.9991	0.0000
Self-Consistency	0.2140	1.0000	3.9991	0.0000
Selective-SC	0.2142	1.0000	3.9991	0.0000
Budgeted-Cascade	0.2140	1.0000	3.9991	0.0000
Few-Shot Exemplar	0.2142	1.0000	3.9991	0.0000

Metric aliases are expanded in the Metric Glossary section.

Table 7: Core results for full_ai2_arc_27b.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	185	2.38	0.3135	1.000	55.925	0.00
Concise-CoT	185	2.38	0.2486	1.000	61.129	0.00
Least-to-Most	185	2.38	0.2541	1.000	59.184	0.00
Constraint-Decompose	185	2.38	0.2541	1.000	70.439	0.00
Self-Verify	183	2.35	0.3005	2.000	113.973	0.00
Critique-Refine	183	2.35	0.3497	3.000	178.838	0.00
Self-Consistency	183	2.35	0.2842	3.000	50.998	0.00
Selective-SC	183	2.35	0.2896	1.077	55.039	0.00
Budgeted-Cascade	183	2.35	0.3115	1.109	0.000	100.00
Few-Shot Exemplar	183	2.35	0.2896	1.000	57.056	0.00

Completed records: 1838 / 77870 (2.36%). Expected examples per strategy: 7787. Model: Qwen3.5-27B-FP8. Best observed strategy so far (by primary): Critique-Refine.

Table 8: Dataset-specific metric means for full_ai2_arc_27b.

Strategy	acc	fmt-valid	choice-count	text-match
Direct	0.3135	1.0000	4.0000	0.0000
Concise-CoT	0.2486	1.0000	4.0000	0.0000
Least-to-Most	0.2541	1.0000	4.0000	0.0000
Constraint-Decompose	0.2541	1.0000	4.0000	0.0000
Self-Verify	0.3005	1.0000	4.0000	0.0000
Critique-Refine	0.3497	1.0000	4.0000	0.0000
Self-Consistency	0.2842	1.0000	4.0000	0.0000
Selective-SC	0.2896	1.0000	4.0000	0.0000
Budgeted-Cascade	0.3115	1.0000	4.0000	0.0000
Few-Shot Exemplar	0.2896	1.0000	4.0000	0.0000

Metric aliases are expanded in the Metric Glossary section.

Table 9: Core results for full_livebench.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	60	4.18	0.0000	1.000	96.214	0.00
Concise-CoT	60	4.18	0.0000	1.000	97.943	0.00
Least-to-Most	60	4.18	0.0000	1.000	93.906	0.00
Constraint-Decompose	60	4.18	0.0000	1.000	92.207	0.00
Self-Verify	60	4.18	0.0000	2.000	182.435	0.00
Critique-Refine	60	4.18	0.0000	3.000	209.885	0.00
Self-Consistency	60	4.18	0.0000	3.000	96.902	0.00
Selective-SC	59	4.11	0.0000	2.254	95.457	0.00
Budgeted-Cascade	60	4.18	0.0000	2.200	4.766	93.33
Few-Shot Exemplar	59	4.11	0.0000	1.000	88.780	0.00

Completed records: 598 / 14360 (4.16%). Expected examples per strategy: 1436. Model: Qwen3.5-27B-FP8. Best observed strategy so far (by primary): Direct.

Table 10: Dataset-specific metric means for full_livebench.

Strategy	proxy-exact	proxy-contains	official-flag
Direct	0.0000	0.0000	1.0000
Concise-CoT	0.0000	0.0000	1.0000
Least-to-Most	0.0000	0.0000	1.0000
Constraint-Decompose	0.0000	0.0000	1.0000
Self-Verify	0.0000	0.0000	1.0000
Critique-Refine	0.0000	0.0000	1.0000
Self-Consistency	0.0000	0.0000	1.0000
Selective-SC	0.0000	0.0000	1.0000
Budgeted-Cascade	0.0000	0.0000	1.0000
Few-Shot Exemplar	0.0000	0.0000	1.0000

Metric aliases are expanded in the Metric Glossary section.

Table 11: Core results for full_room_assignment.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	359	0.95	0.0232	1.000	67.592	0.00
Concise-CoT	359	0.95	0.0120	1.000	66.666	0.00
Least-to-Most	359	0.95	0.0098	1.000	66.316	0.00
Constraint-Decompose	358	0.94	0.0132	1.000	65.734	0.00
Self-Verify	358	0.94	0.0997	2.000	134.012	0.00
Critique-Refine	358	0.94	0.0493	3.000	171.969	0.00
Self-Consistency	358	0.94	0.0307	3.000	66.713	0.00
Selective-SC	358	0.94	0.0241	2.547	68.344	0.00
Budgeted-Cascade	358	0.94	0.0261	1.061	3.786	96.93
Few-Shot Exemplar	358	0.94	0.0218	1.000	65.213	0.00

Completed records: 3583 / 379840 (0.94%). Expected examples per strategy: 37984. Model: Qwen3.5-4B. Best observed strategy so far (by primary): Self-Verify.

Table 12: Dataset-specific metric means for full_room_assignment.

Strategy	room-exact	entity-room	all-rooms	fmt-valid	pred-rooms
Direct	0.0179	0.0286	0.0056	0.9499	9.9443
Concise-CoT	0.0074	0.0166	0.0028	0.9053	8.7632
Least-to-Most	0.0060	0.0136	0.0028	0.9081	8.6045
Constraint-Decompose	0.0090	0.0175	0.0000	0.8659	8.0922
Self-Verify	0.0919	0.1075	0.0726	0.9441	12.5838
Critique-Refine	0.0439	0.0547	0.0307	0.9944	12.5866
Self-Consistency	0.0280	0.0335	0.0140	0.9413	11.1006
Selective-SC	0.0212	0.0270	0.0112	0.9274	10.9972
Budgeted-Cascade	0.0207	0.0315	0.0084	0.9525	10.0223
Few-Shot Exemplar	0.0170	0.0265	0.0028	0.9749	8.1844

Metric aliases are expanded in the Metric Glossary section.

Table 13: Core results for full_room_assignment_27b.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	35	0.09	0.4732	1.000	95.236	0.00
Concise-CoT	35	0.09	0.2581	1.000	101.051	0.00
Least-to-Most	35	0.09	0.1914	1.000	105.805	0.00
Constraint-Decompose	34	0.09	0.1332	1.000	100.735	0.00
Self-Verify	34	0.09	0.4831	2.000	176.307	0.00
Critique-Refine	34	0.09	0.2283	3.000	237.385	0.00
Self-Consistency	34	0.09	0.4026	3.000	100.384	0.00
Selective-SC	34	0.09	0.3761	1.882	98.698	0.00
Budgeted-Cascade	34	0.09	0.4857	1.059	0.000	100.00
Few-Shot Exemplar	34	0.09	0.4559	1.000	98.386	0.00

Completed records: 343 / 379840 (0.09%). Expected examples per strategy: 37984. Model: Qwen3.5-27B-FP8. Best observed strategy so far (by primary): Budgeted-Cascade.

Table 14: Dataset-specific metric means for full_room_assignment_27b.

Strategy	room-exact	entity-room	all-rooms	fmt-valid	pred-rooms
Direct	0.4643	0.4821	0.4286	1.0000	15.4857
Concise-CoT	0.2567	0.2595	0.2000	1.0000	15.1429
Least-to-Most	0.1914	0.1914	0.1714	1.0000	16.5429
Constraint-Decompose	0.1299	0.1365	0.0882	1.0000	15.2941
Self-Verify	0.5059	0.4603	0.4706	1.0000	10.8529
Critique-Refine	0.2088	0.2478	0.1471	1.0000	9.9118
Self-Consistency	0.3971	0.4081	0.3529	1.0000	15.2647
Selective-SC	0.3618	0.3904	0.2647	1.0000	15.5588
Budgeted-Cascade	0.4779	0.4934	0.4412	1.0000	15.2647
Few-Shot Exemplar	0.4632	0.4485	0.4412	0.9706	14.7647

Metric aliases are expanded in the Metric Glossary section.

Table 15: Core results for full_truthfulqa.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	254	31.09	0.0024	1.000	41.948	6.30
Concise-CoT	254	31.09	0.0018	1.000	48.023	7.48
Least-to-Most	253	30.97	0.0015	1.000	53.313	7.51
Constraint-Decompose	253	30.97	0.0013	1.000	56.686	7.91
Self-Verify	253	30.97	0.0024	2.000	87.245	6.32
Critique-Refine	253	30.97	0.0019	3.000	140.701	0.00
Self-Consistency	253	30.97	0.0022	3.000	48.962	0.00
Selective-SC	253	30.97	0.0024	3.000	57.288	0.00
Budgeted-Cascade	253	30.97	0.0025	3.000	31.350	41.11
Few-Shot Exemplar	253	30.97	0.0020	1.000	43.946	6.32

Completed records: 2532 / 8170 (30.99%). Expected examples per strategy: 817. Model: Qwen3.5-4B. Best observed strategy so far (by primary): Budgeted-Cascade.

Table 16: Dataset-specific metric means for full_truthfulqa.

Strategy	truth-delta	correct-sim	incorrect-sim	best-sim	info-proxy	uninformative
Direct	0.0024	0.0188	0.0164	0.0140	1.0000	0.0000
Concise-CoT	0.0018	0.0157	0.0139	0.0117	1.0000	0.0000
Least-to-Most	0.0015	0.0134	0.0119	0.0101	1.0000	0.0000
Constraint-Decompose	0.0013	0.0117	0.0104	0.0089	1.0000	0.0000
Self-Verify	0.0024	0.0221	0.0196	0.0169	1.0000	0.0000
Critique-Refine	0.0019	0.0158	0.0139	0.0120	1.0000	0.0000
Self-Consistency	0.0022	0.0180	0.0158	0.0137	1.0000	0.0000
Selective-SC	0.0024	0.0188	0.0164	0.0143	1.0000	0.0000
Budgeted-Cascade	0.0025	0.0239	0.0213	0.0179	1.0000	0.0000
Few-Shot Exemplar	0.0020	0.0179	0.0160	0.0134	1.0000	0.0000

Metric aliases are expanded in the Metric Glossary section.

Table 17: Core results for full_truthfulqa_27b.

Strategy	n	cov%	primary	api	time(s)	cache%
Direct	76	9.30	0.0044	1.000	38.026	5.26
Concise-CoT	76	9.30	0.0065	1.000	37.572	3.95
Least-to-Most	76	9.30	0.0055	1.000	60.796	5.26
Constraint-Decompose	74	9.06	0.0019	1.000	70.322	5.41
Self-Verify	75	9.18	0.0095	2.000	92.957	2.67
Critique-Refine	75	9.18	0.0091	3.000	133.494	0.00
Self-Consistency	75	9.18	0.0043	3.000	39.073	0.00
Selective-SC	75	9.18	0.0043	2.840	39.121	0.00
Budgeted-Cascade	75	9.18	0.0112	2.840	0.002	100.00
Few-Shot Exemplar	75	9.18	0.0048	1.000	32.544	5.33

Completed records: 752 / 8170 (9.20%). Expected examples per strategy: 817. Model: Qwen3.5-27B-FP8. Best observed strategy so far (by primary): Budgeted-Cascade.

Table 18: Dataset-specific metric means for full_truthfulqa_27b.

Strategy	truth-delta	correct-sim	incorrect-sim	best-sim	info-proxy	uninformative
Direct	0.0044	0.0389	0.0345	0.0287	1.0000	0.0000
Concise-CoT	0.0065	0.0514	0.0449	0.0363	0.9879	0.0000
Least-to-Most	0.0055	0.0333	0.0277	0.0247	1.0000	0.0000
Constraint-Decompose	0.0019	0.0235	0.0216	0.0183	1.0000	0.0000
Self-Verify	0.0095	0.0687	0.0592	0.0514	1.0000	0.0000
Critique-Refine	0.0091	0.0872	0.0781	0.0682	1.0000	0.0000
Self-Consistency	0.0043	0.0397	0.0354	0.0287	1.0000	0.0000
Selective-SC	0.0043	0.0450	0.0407	0.0340	1.0000	0.0000
Budgeted-Cascade	0.0112	0.0651	0.0540	0.0490	1.0000	0.0000
Few-Shot Exemplar	0.0048	0.0482	0.0434	0.0333	1.0000	0.0000

Metric aliases are expanded in the Metric Glossary section.

Table 19: Execution log summary by session (derived from run log files).

Session	Attempts	Finished	Success	Failed	LastExit	LatestLog
rb_ai2_arc	1	0	0	0	–	run_20260401_181910.log
rb_ai2_arc_27b	2	1	0	1	139	run_20260401_182217.log
rb_livebench	4	1	1	0	0	run_20260401_175234.log
rb_room_assignment	4	1	1	0	0	run_20260401_175234.log
rb_room_assignment_27b	1	0	0	0	–	run_20260401_182155.log
rb_truthfulqa	44	40	1	39	139	run_20260402_055318.log
rb_truthfulqa_27b	13	12	0	12	139	run_20260402_094739.log